Tomohiko Nakamura

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Research Interests

Signal-processing-inspired deep learning, audio and music signal processing, and machine learning

Job

Senior Researcher **Apr. 2023–Present**
Intelligent Media Processing Research Team, Artificial Intelligence Research Center, National Institute of Advanced Industrial Science and Technology (AIST)

Visiting Researcher **Nov. 2025–Present**
Graduate School of Information Science and Technology, The University of Tokyo

Project Research Associate **Sep. 2019–Mar. 2023**
Graduate School of Information Science and Technology, The University of Tokyo

Researcher **Apr. 2016–Aug. 2019**
Intelligent Systems Laboratory, SECOM

Research Fellow (DC2) **Apr. 2015–Mar. 2016**
Japan Society for the Promotion of Science (JSPS)

Education

Ph.D. degree in Information Science and Technology **Mar. 2016**
Graduate School of Information Science and Technology, The University of Tokyo

Master's degree in Information Science and Technology **Mar. 2013**
Graduate School of Information Science and Technology, The University of Tokyo

Bachelor's degree in Engineering **Mar. 2011**
Faculty of Engineering, The University of Tokyo

Teaching

Advanced Signal Processing **June 2020, 2022, 2024, and 2026**
Graduate School of Information Science and Technology, The University of Tokyo

Applied Gaussian Process and Machine Learning **Dec. 2021**
Graduate School of Information Science and Technology, The University of Tokyo

Keisu Student Experiment **Apr. 2020–Mar. 2023**
Department of Mathematical Engineering and Information Physics, The University of Tokyo

Languages

Japanese (native), English (basic)

Competitive Funds

JST PRESTO (Fundamental Innovation for Real-World Intelligent Systems)
Variable time-resolution audio processing framework adaptive to acoustic scenes *Oct. 2025–Mar. 2029*

JSPS KAKENHI Grant-in-Aid for Scientific Research (B)
Development of deep-layered analysis-by-synthesis techniques for acoustic scene analysis with human intervention *Apr. 2023–Mar. 2027*

JST ACT-X (Frontier of Mathematics and Information Science)

Sampling-frequency-independent deep learning for audio media processing *Oct. 2021–Mar. 2025*

+ 4 funds as representative, 4 funds as co-researcher, and 2 funds as research participant.

Publications

Journal Papers

[1] Yoshiaki Bando, Tomohiko Nakamura, Satoru Fukayama, and Shinji Watanabe, "Online frontend system for multi-talker DSR using neural blind source separation and diarization," *IEEE Transactions on Audio, Speech and Language Processing*, vol. 34, pp. 3698–3713, July 2026.

[2] Ren Uchida, Kohei Yatabe, and Tomohiko Nakamura, "Encoder-masking-decoder networks using orthogonal convolutional layer as invertible linear encoder," *Acoustical Science and Technology*, vol. advpub, no. e26.10, May 2026.

[3] Kanami Imamura, Tomohiko Nakamura, Norihiro Takamune, Kohei Yatabe, and Hiroshi Saruwatari, "Stride conversion algorithms for convolutional layers and its application to sampling-frequency-independent deep neural networks," *Signal Processing*, vol. 242, Nov. 2025.

[4] Yuki Ito, Tomohiko Nakamura, Shoichi Koyama, Shuichi Sakamoto, and Hiroshi Saruwatari, "Spatial upsampling of head-related transfer function using neural network conditioned on source position and frequency," *IEEE Open Journal of Signal Processing*, vol. 6, pp. 1109–1123, Sept. 2025.

[5] +11 papers

Peer-Reviewed International Conferences and Workshops

[1] Keisuke Toyoshima, Ayaka Yomoda, Yukihiro Kida, Mitsuyasu Deguchi, Yoshitaka Watanabe, Tomohiko Nakamura, Kazuki Maruta, and Daisuke Hisano, "At-sea evaluation of discretely-modulated deep joint source-channel coding for underwater acoustic communications," in *Proceedings of International Conference on Underwater Networks and Systems*, 2026.

[2] Kengo Takemoto, Tomohiko Nakamura, and Hiroshi Saruwatari, "Differentiable digital signal processing mixture model-based diffusion for synthesis parameter estimation from harmonic sound mixtures," in *Proceedings of Asia Pacific Signal and Information Processing Association Annual Summit and Conference*, Nov. 2026.

[3] Yuma Narahata, Tomohiko Nakamura, Yuki Saito, and Hiroshi Saruwatari, "Lead vocal separation from vocal ensemble mixtures using phoneme alignment," in *Proceedings of Asia Pacific Signal and Information Processing Association Annual Summit and Conference*, Nov. 2026.

[4] +48 papers

Invited Talks

[1] Tomohiko Nakamura, "Trends and prospects for audio source separation using deep learning," *Meeting on Technical Committee on Engineering Acoustics, IEICE*, vol. 124, no. 389, EA2024-93, p. 104, Mar. 2025. (in Japanese)

[2] Daichi Kitamura and Tomohiko Nakamura, "Fundamentals and applications of audio source separation — A guide to becoming an expert," *2023 Otogaku Symposium*, vol. 2023-MUS-137 no. 35, Jun. 2023. (in Japanese)

[3] Tomohiko Nakamura, "Signal-processing-inspired deep learning," *IEEE NZ Signal Processing/Information Theory Joint Chapter, co-hosted by the Acoustics Research Centre, University of Auckland*, Dec. 2022.

[4] +1 invited presentations

Overview Papers

1. Shoichi Koyama, Juliano Ribeiro, Tomohiko Nakamura, Natsuki Ueno, and Mirco Pezzoli, "Physics-informed machine learning for sound field estimation," *Special Issue on Model-Based and Data-Driven Audio Signal Processing, IEEE Signal Processing Magazine*, vol. 41, pp. 60–71, Nov. 2024.

Patents

[1] Tomohiko Nakamura, "Object recognition device, method, and program," Japan Patent JP7349288, 13-Sept-2023.

[2] +10 patents

Awards

1. The Awaya Kiyoshi Research Award, ASJ, Mar. 2024.
2. The Itakura Prize Innovative Young Researcher Award, ASJ, Mar. 2022.
3. +12 awards